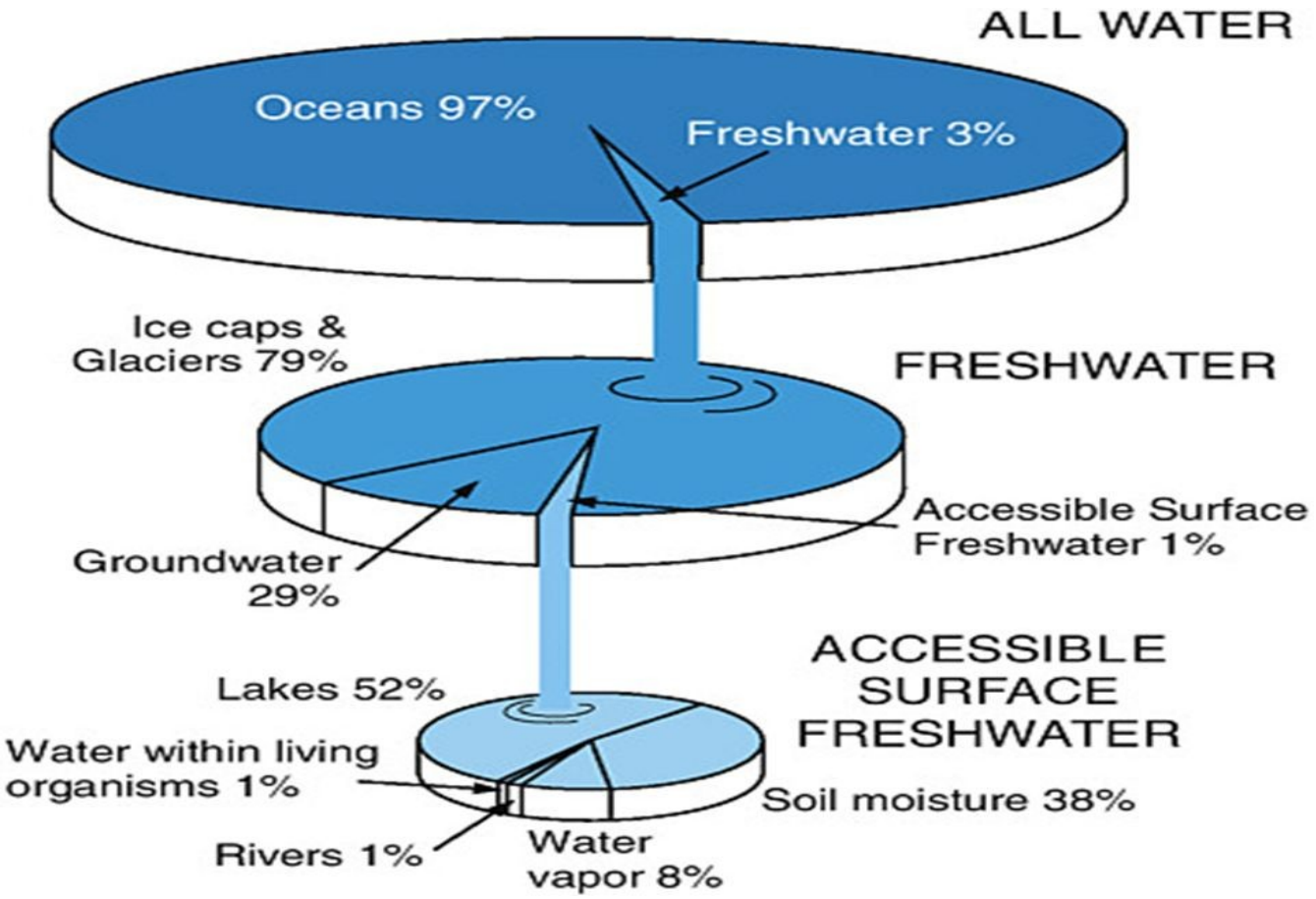
Water Resource



Water Resource

Distribution of Global Water



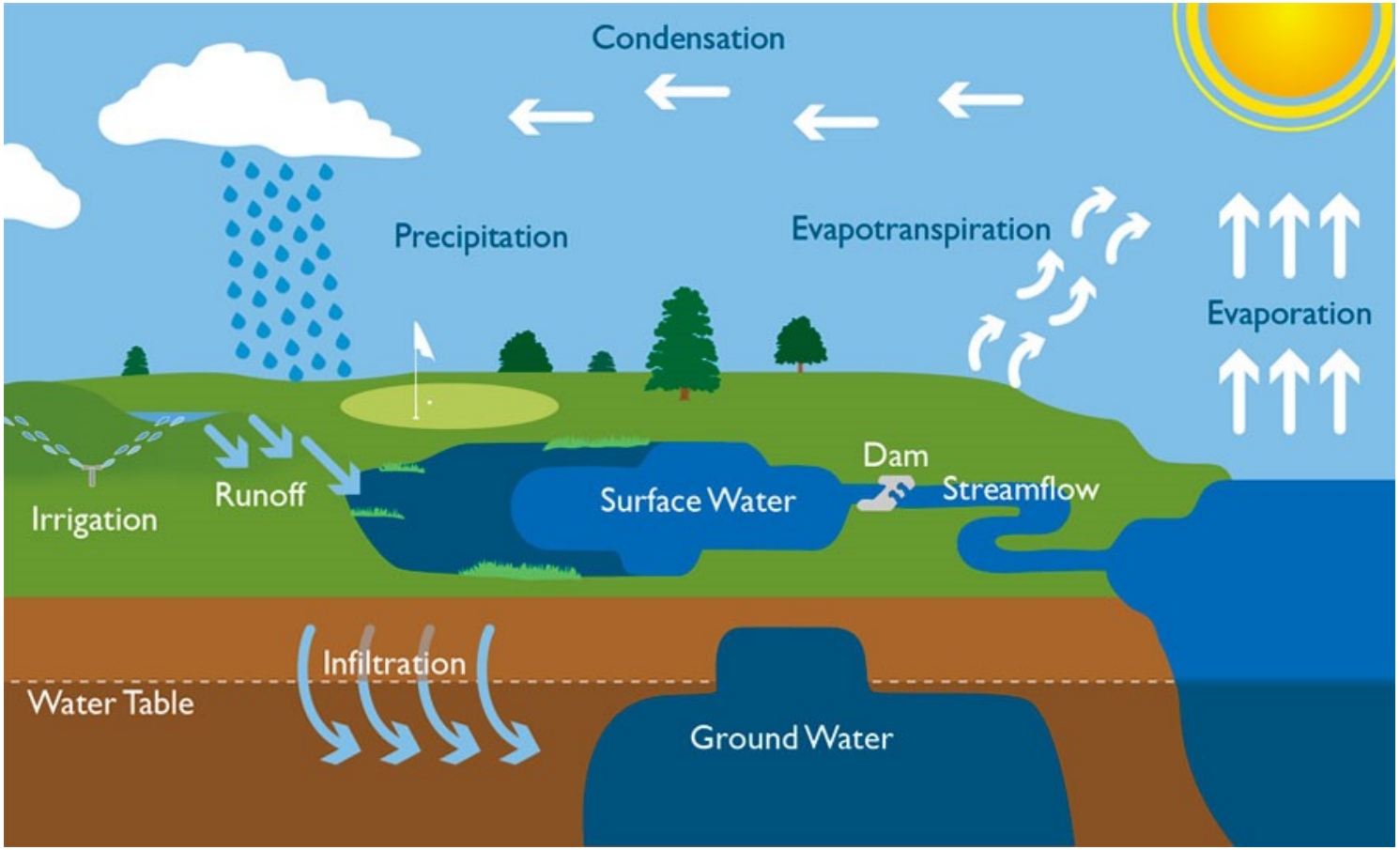


Water Resource

Hydrological Cycle!!!

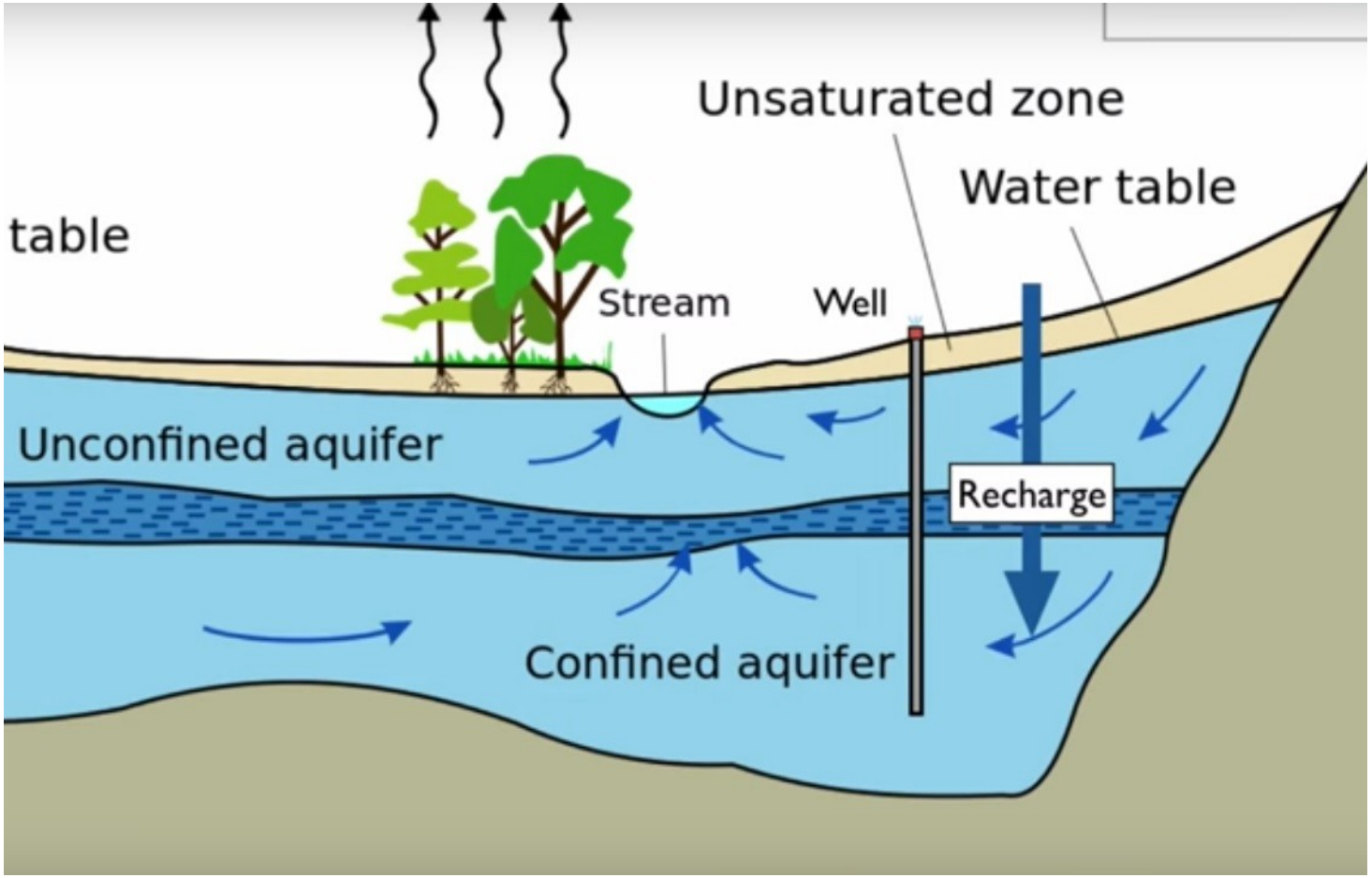
Water Resource

Hydrological Cycle



Water Resource

Fresh water



Water Resource

Water Use

Agriculture – 70%
Industry – 22%
Domestic – 8%

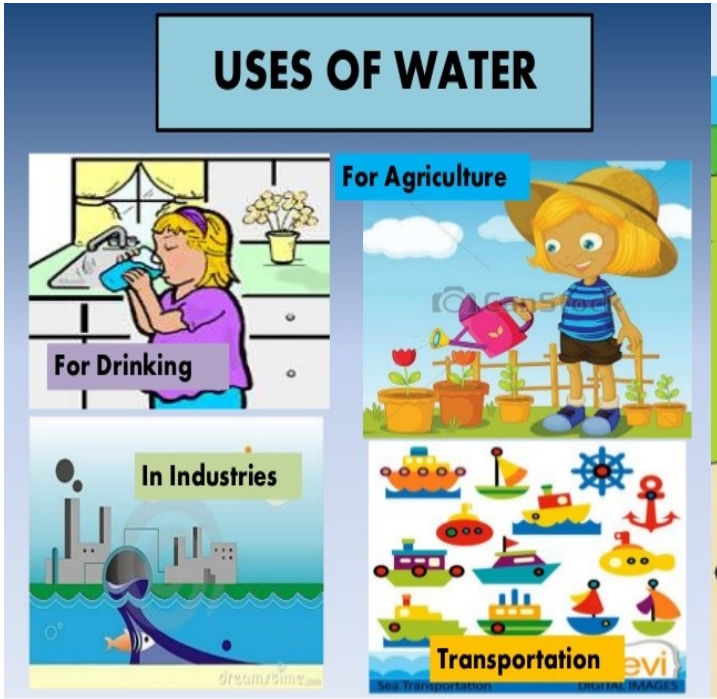
Water Storage Ground water – Aquifer

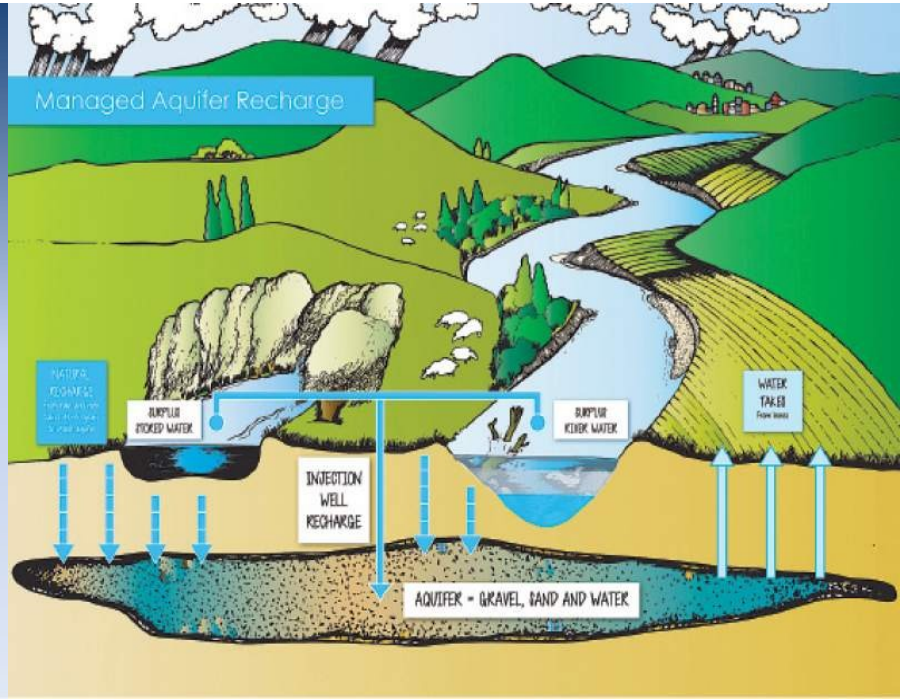
Helps!!

Potable water to drink
Reduce, Reuse, and Recycle (3R)

Concern!!

Ground water depletion
Untreated ground water





Water Resource

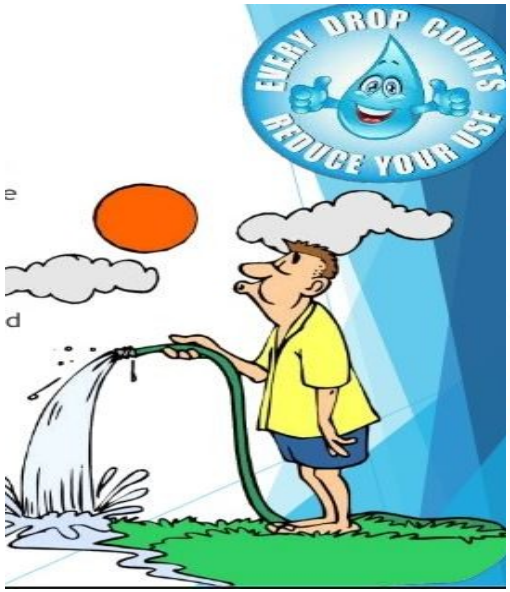
Water Storage – Surface water - Reservoir

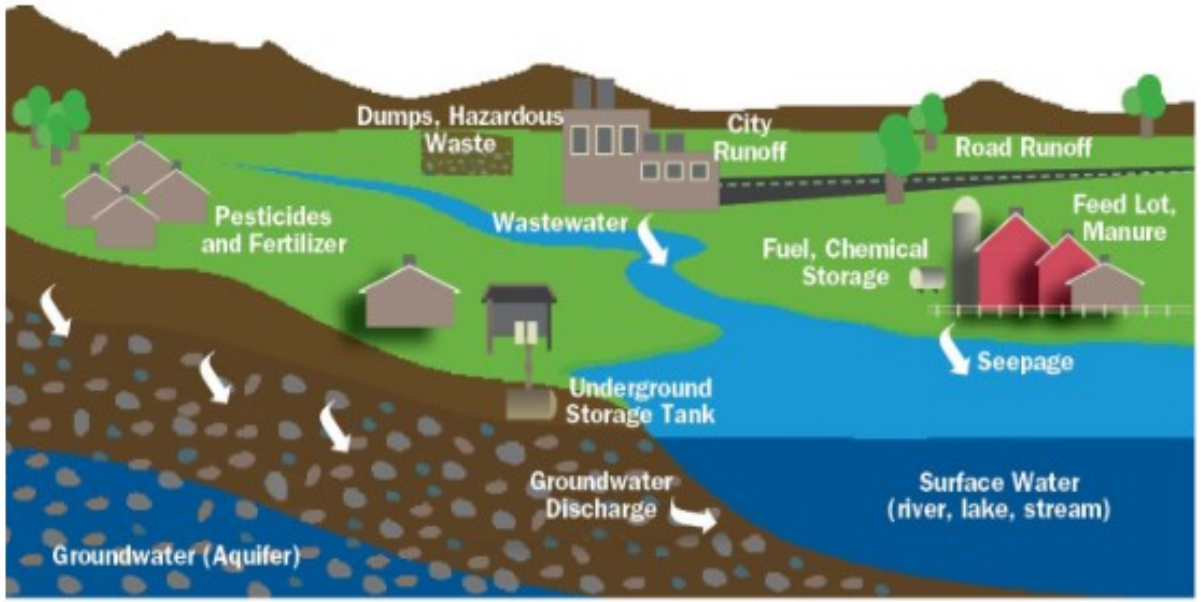
Helps!!

Energy Production
Reduce Flood
Irrigation
Reclaimed water

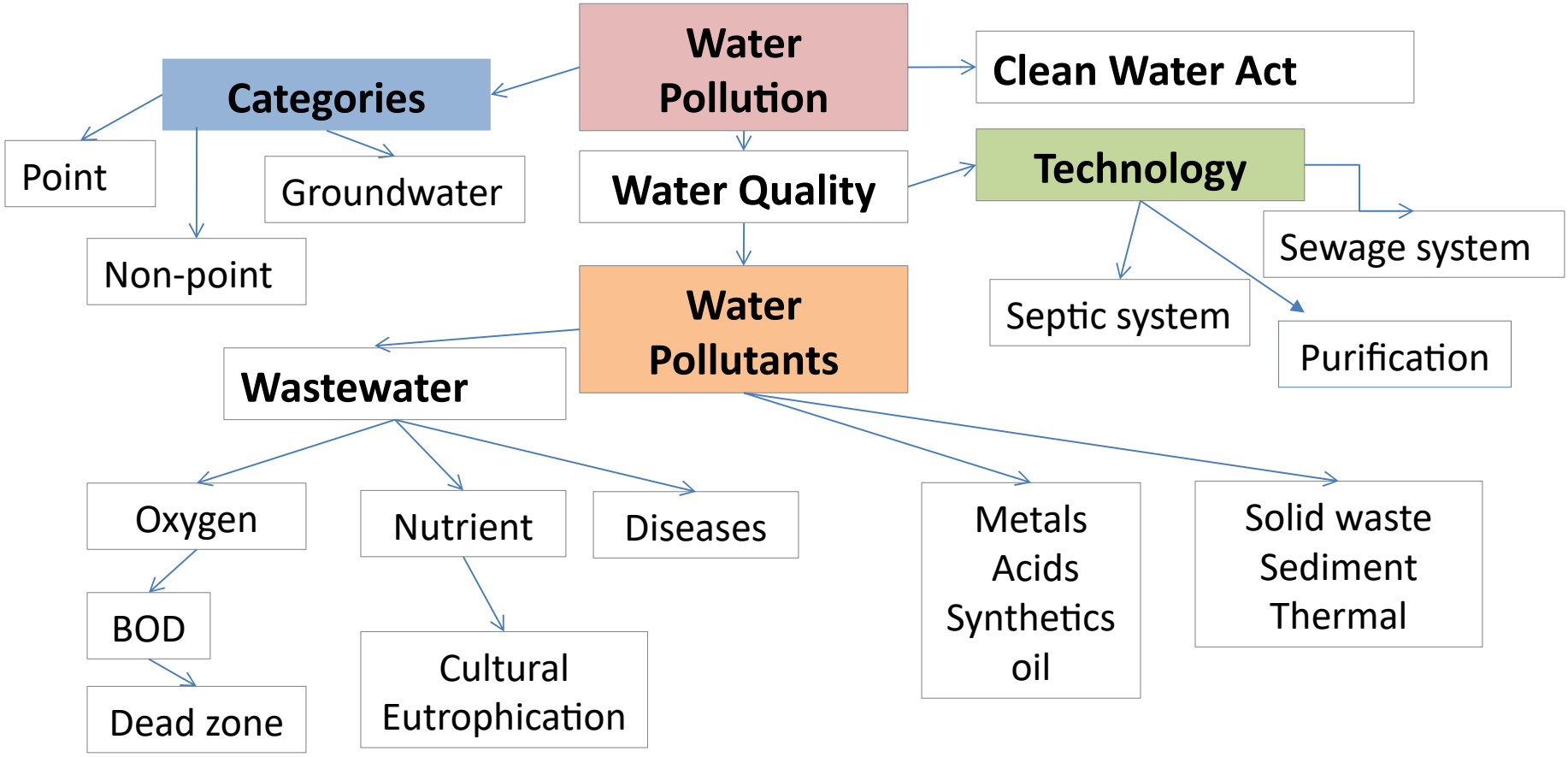
Concern!!

Destruction
Less Evaporation
Wildlife hampered
Eviction of people or community





Water Resource



Water Resource

Water Pollution

The contamination of surface water (streams, rivers , lakes, oceans) or groundwater with substances produced through human activities and natural phenomenon that negatively affect organisms.

Pollution Categories:

Point sources – distinct locations that pump waste or contaminants into a waterway. You can target (or point to).

Non-point sources – diffuse areas such as an entire farming region that pollutes a waterway. Hard to target where the source exactly is.

Groundwater pollution/depletion - long-term **water**-level/quality declines caused by sustained **groundwater** pumping a/o contamination

Water Resource

Categories



Point



Groundwater



Non-point

Water Resource

Water pollution

Causes

Sewage
Nutrients
Waste water
Chemical waste
Radioactive waste
Oil pollution



Impacts

Death of aquatic (water) animals
Disruption of food-chains
Diseases
Destruction of ecosystems



Water Resource

Water quality

"**Water quality**" is a term used here to express the suitability of water to sustain various uses or processes. Any particular use will have certain requirements for the physical, chemical or biological characteristics of water (UNEP/WHO, 1996)

Water quality – parameters:

- **Physical parameters**. Colour, odour, turbidity, taste, temperature and electrical conductivity
- **Chemical parameters.** the amount of carbonates, sulphates, chlorides, fluorides, nitrates, and metal ions
- **Biological parameters.** Quantity Existence of algae, fungi, viruses, protozoa and bacteria

Water Resource

Water quality and dissolved oxygen (DO) content in parts per million (ppm) at 20°C

BOD Level in mg/liter	Water Quality
1 - 2	**Very Good:** There will not be much organic matter present in the water supply.
3 - 5	**Fair:** Moderately Clean
6 - 9	**Poor:** Somewhat Polluted - Usually indicates that organic matter present and microorganisms are decomposing that waste.
100 or more	**Very Poor:** Very Polluted - Contains organic matter.

- Only a few fish species can survive in the water less than 4ppm at 20°C

Water Resource

Water Pollutants

Types:

(i) Sewage Pollutants (Domestic and Municipal Waste)
(ii) Industrial Pollutants
(iii) Agricultural Pollutants
(iv) Radioactive and Thermal Pollutants

In the form of:

- Human Wastewater
- Chemical pollutants
- Non-chemical pollutants



Water Resource

Human Wastewater

Three reasons scientists are concerned about **waste water**

1. Oxygen demanding waste like bacteria (E.coli) that put a large demand for oxygen in the water
2. Nutrients that area released from wastewater decomposition can make the water more fertile causing eutrophication
3. Wastewater can carry a wide variety of disease-causing organism.



Water Resource

Biochemical Oxygen Demand (BOD)

BOD – the amount of oxygen a quantity of water uses over a period of time at a specific temperature

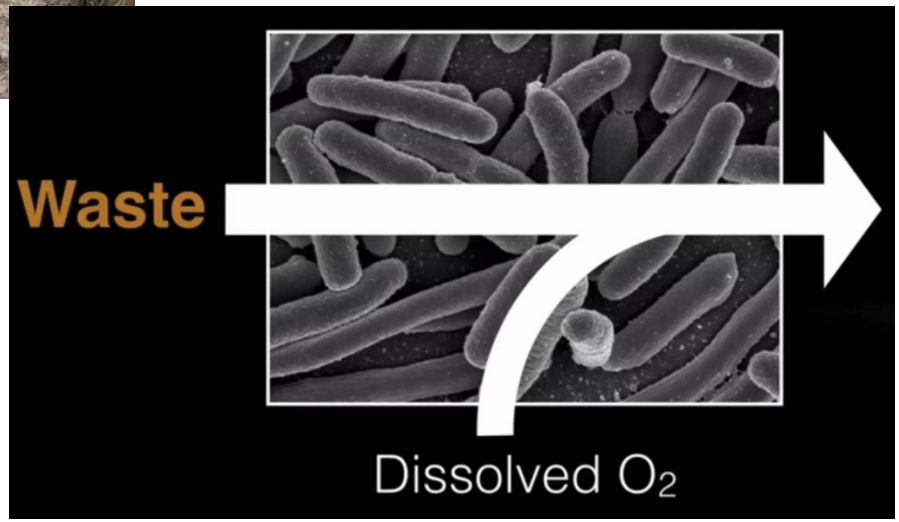
- Lower BOD values indicate the water is less polluted and higher BOD values indicate it is more polluted by wastewater
- When bodies of water have high oxygen demand due to microbial decomposition, the amount of O_2 for other organisms can be very low

Water Resource

Wastewater



Decomposers





Dead zone



Water Resource

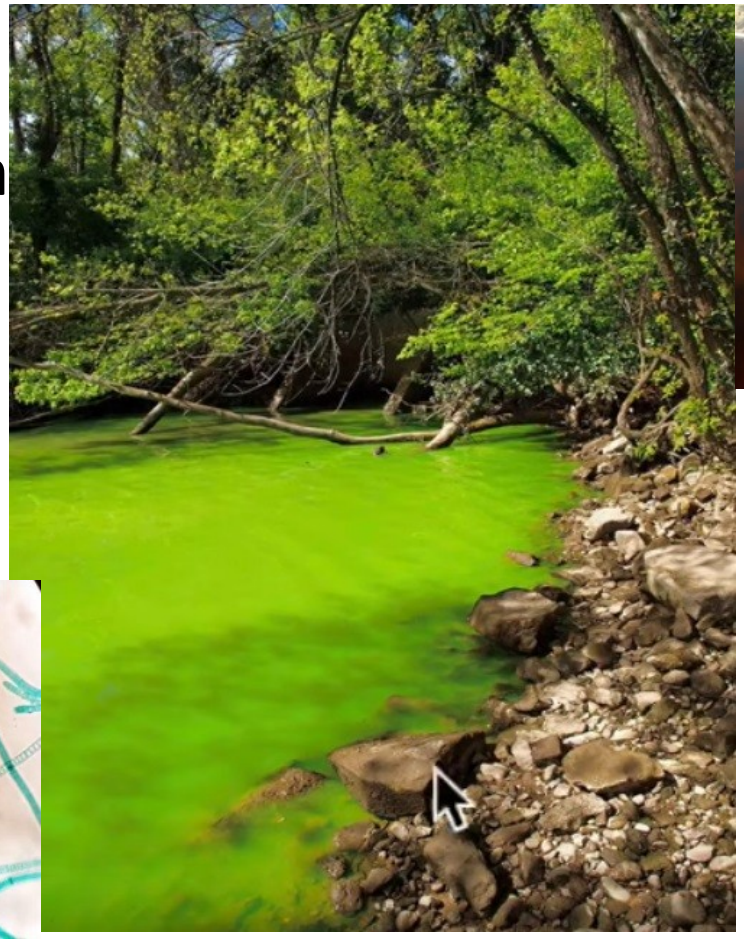
Eutrophication

Eutrophication is an abundance of fertility to a body of water. For instance, Algae bloom

- caused by an increase in nutrients (including nitrogen and phosphorus), such as fertilizers.
- can cause a rapid growth of algae which eventually dies, causing the microbes to increase the BOD.

Water Resource

Cultural Eutrophication





Algae



Dead zone

Water Resource

diseases

Common diseases from Human Wastewater

- ✓ Cholera
- ✓ Typhoid fever
- ✓ Stomach flu
- ✓ Diarrhea
- ✓ Hepatitis

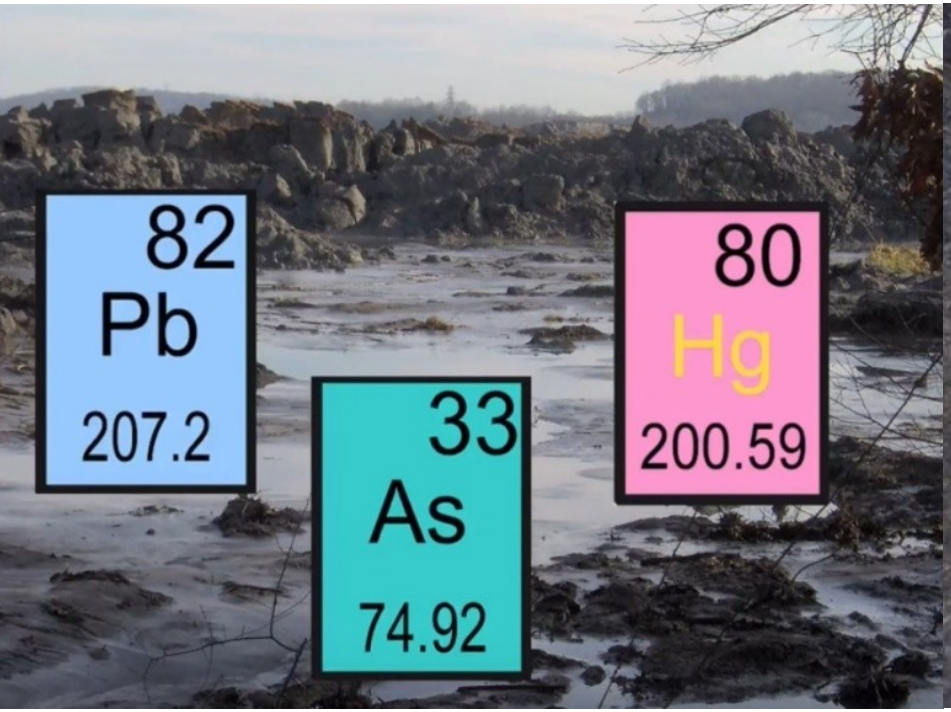
Facts!

1/6 of the world's population does not have access to sufficient supplies of drinking water

½ of the diarrheal and malaria deaths could be prevented with safe drinking water, and proper sanitation and hygiene

Water Resource

Chemical Pollutants

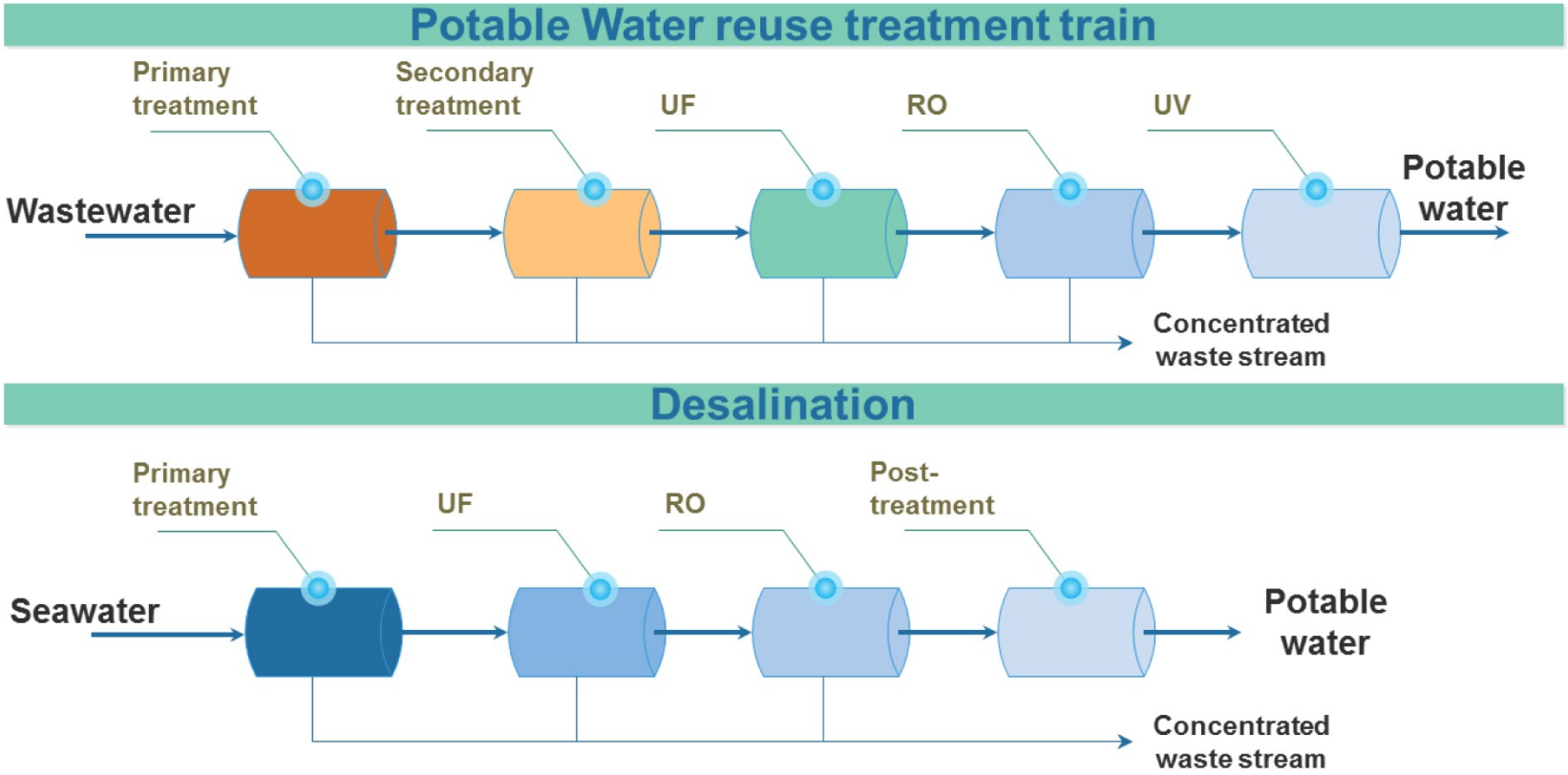


Non-chemical Pollutants



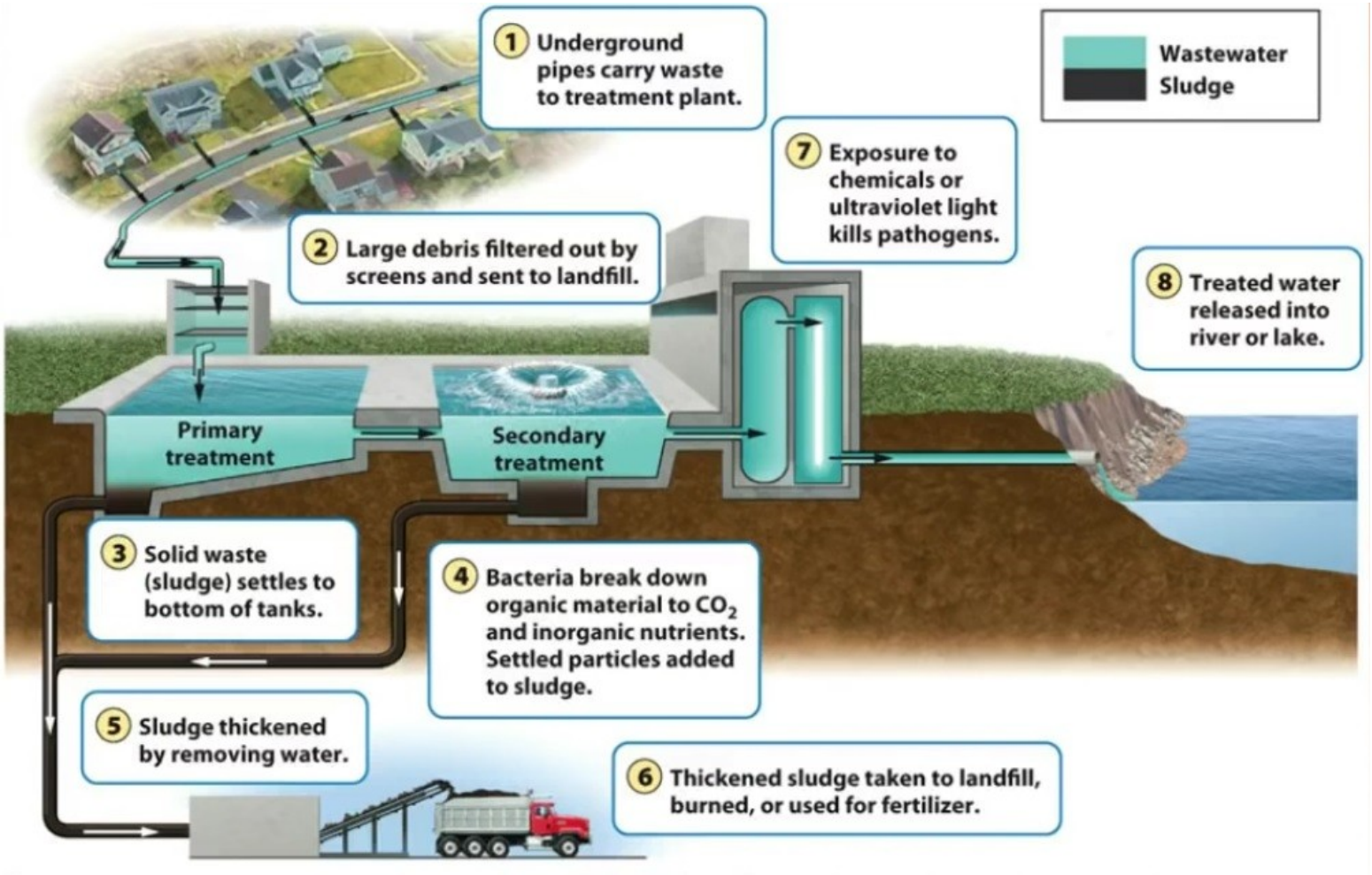
Water Resource

Technology Used to Water Treatment



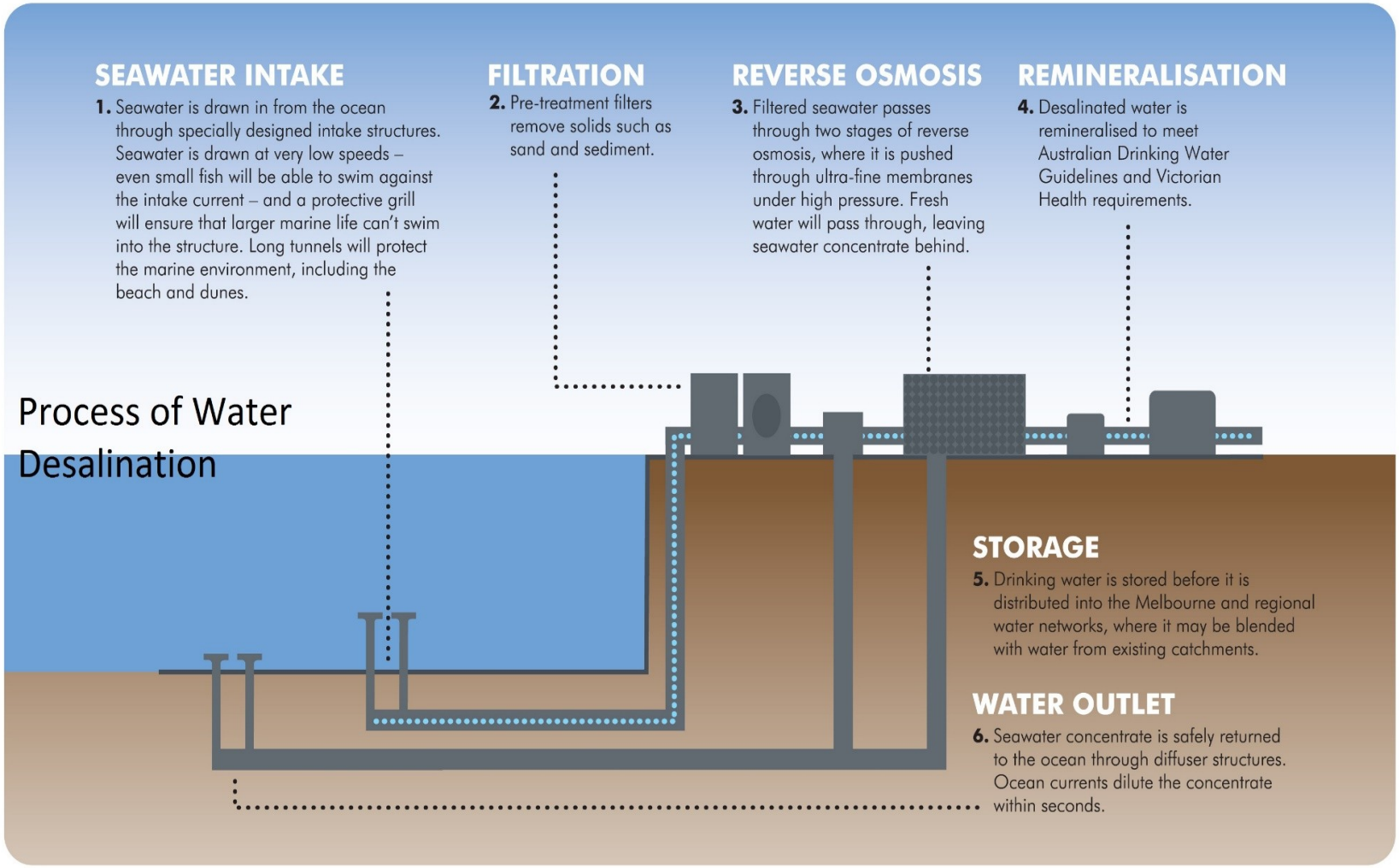
Water Resource

Sewage Treatment Plant



Water Resource

Sewage Treatment Plant



Water Resource

Treatments for Human and Animal Wastewater

- **Septic systems-** a large container that receives wastewater from the house. Anything that floats to the top is "scum." Anything that sinks is "sludge." The middle layer is clear water called "septage." The septage contains bacteria, but also pathogenic organisms, N and P. The water flows through pipes and lawn called the leach field.

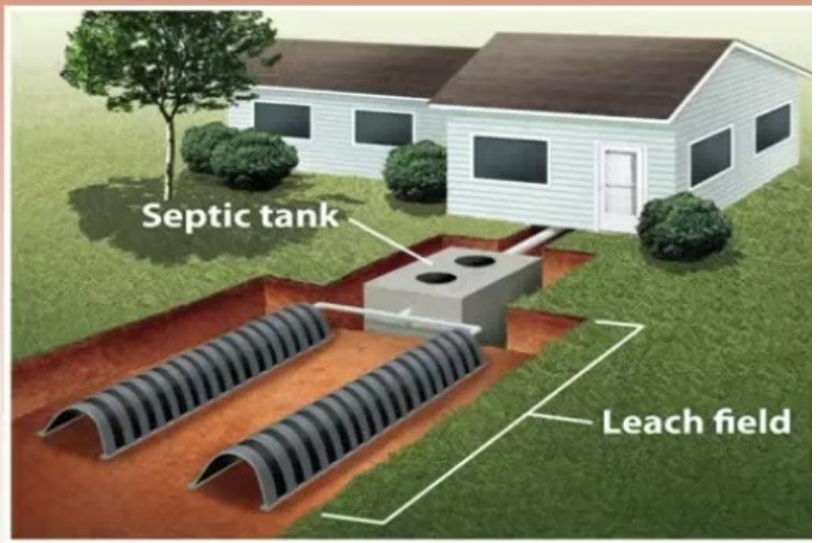


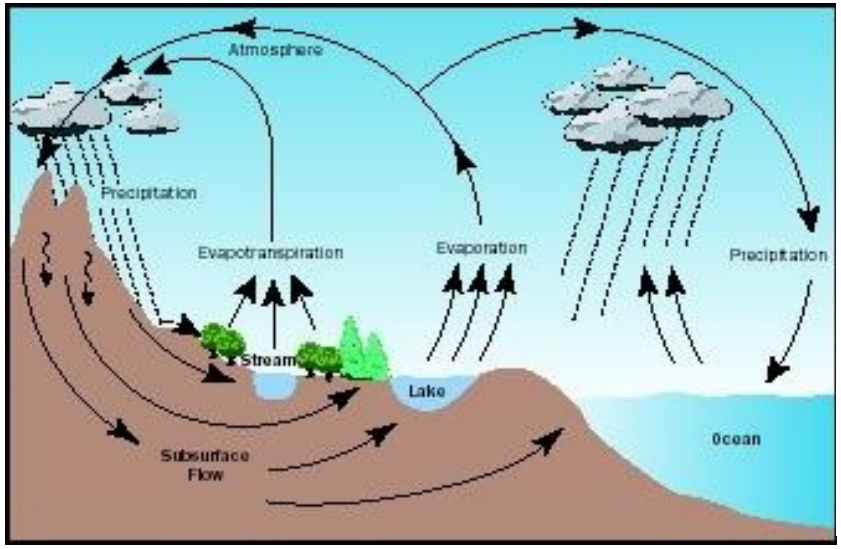
Figure 14.5
Environmental Science
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Pro - No electricity is used; con – must be pumped out every 5-10 years

Water Resource

Water Movement

Ground water Recharge – Natural
Saturated flow
Irrigation
Potable water supply
Desalination



Water Conservation

- low cost water through subsidies incentives
-Water-saving Devices







Thank you for your attention

Welcoming Q & A !!